

## A Delay System Method for Designing Event-Triggered Controllers of Networked Control Systems

Dong Yue, Engang Tian, and Qing-Long Han

**Abstract**—This note is concerned with event-triggered  $H_\infty$  controller design for networked control systems. A novel event-triggering scheme is proposed, which has some advantages over some existing schemes. A delay system model for the analysis is firstly constructed by investigating the effect of the network transmission delay. Then, based on this model, criteria for stability with an  $H_\infty$  norm bound and criteria for co-designing both the feedback gain and the trigger parameters are derived. These criteria are formulated in terms of linear matrix inequalities. Simulation results have shown that the proposed event-triggering scheme is superior to some existing event-triggering schemes in the literature.

**Index Terms**—Event-triggered control, Lyapunov functional, network-induced delay, networked control systems.

### I. INTRODUCTION

Networked control systems (NCSs) have a relatively new structure where sensors, controllers and plants are often connected over a common network medium [8], [18], [24], [30]. Because of the insertion of the network, the tasks in traditional systems, such as the control problem and signal estimation problem, should be re-considered. In the past decade, the control design problem for NCSs has experienced an increasing attention in the literature [5], [6], [23], [26], [28]. In these works, most results use a periodic triggered control method (also called a time-triggered control) for system modeling and analysis due to the easy implementation and analysis. In this triggering method, a fixed sampling interval should be selected to guarantee a desired performance under worst conditions such as external disturbances, uncertainties, time-delays and so on. However, in practical systems, the worst cases are seldom encountered. Therefore, this kind of triggering method will lead to the sending of many “unnecessary” sampling signals through the network, which will cause high utilization of the communication bandwidth. This kind of over-occupation may have slight impact on point-to-point connected systems, but will affect

networked control systems greatly because the bandwidth of an NCS is often limited.

In [14], Seto *et al.* (1996) addressed how to reduce communication requirements on networked control systems, where the maximum time allowed to elapse was obtained to guarantee the stability of the system. However, the approach in [14] leads to an inherently periodic transmission. Recently, event-triggered techniques for control design, advocating the use of actuation only when some function of the system state exceeds a threshold, have been paid an increased attention in the literature [4], [11], [15], [16]. Event-triggering provides a useful way of determining when the sampling action is carried out, which guarantees that only really “necessary” state signal will be sent out to the controller. Thus, the amount of the sent state signals is relatively little. Compared with periodic sampling methods, the event-triggering method has the following advantages: 1) closer in nature to the way the human behaviour as a controller [2], which only samples when necessary; 2) reduction in the release times of the sensor and then the burden of the network communication; and 3) reduction in the computation cost of the controller and the occupation of the sensor and actuator.

In [10], the event-triggered control with constraint  $\|x(t) - x(t_k)\| \leq \bar{\epsilon}$  was proposed for linear systems with an external disturbance, where the threshold  $\bar{\epsilon}$  of the event generator is a constant. The criteria to guarantee the uniform boundedness of the system were derived. The methods for design or implementation of controllers in the event-triggered form based on dissipation inequalities were proposed for both linear and nonlinear systems [16], [19]. From [16], [19], one can see that the constructed event-triggers rely on continuous supervision of the system state in order to detect whether the current state exceeds the triggering threshold. To implement such a kind of event triggers, the existing system should be retrofitted, which can lead to the complexity of realization of the event-triggered scheme [22]. In these cases, a self-triggered scheme, in which release times are determined by a software approach, was recently proposed [9], [12], [13], [21], [22]. Compared with the event-triggered scheme in [10], [15]–[17], [19]–[21], the self-triggered scheme can provide additional energy savings for the sensor and also less complexity in the implementation. However, as shown in [1], [11], [13], [21], [22], the average release period based on a self-triggered scheme is often smaller than the one based on an event-triggered scheme. Moreover, more constraints on the system structure are needed for design or implementation of controllers in the self-triggered form.

In this note, we focus on event-triggered  $H_\infty$  control for networked systems. The event generator is positioned between the sensor and the controller, which is used to determine whether the newly sampled state should be sent out to the controller. Our event generator is also implemented based on dissipation inequalities as in [10], [15]–[17], [19]–[21]. However, unlike the cases in [10], [15]–[17], [19]–[21], the implementation of our event generator only needs a supervision of the system state in discrete instants. Similar to the self-triggered scheme in [21], no work is needed to retrofit the existing system by using our method. Therefore, it is relatively easier to implement our event-triggered scheme than some existing ones in the literature. Under the event-triggering scheme, a novel model is firstly proposed for the use of system analysis and control design. In the model, the effect of the network transmission delay and the properties of the event-triggering scheme are taken into account. Based on the model, criteria for the stability with an  $H_\infty$  norm bound and controller design are derived, which are expressed in the form of linear matrix inequality. The criteria also establish the relationship between the parameters of the event-trigger, the transmission delay and the feedback gain. Therefore, co-design for both the controller and the parameters of triggers can be carried out by using our method. It should be noted that, some existing [10], [12],

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[20]–[22] can only provide an implementation of an event-triggering scheme or a self-triggering scheme under a given feedback gain. No method was provided in [10], [12], [20]–[22] for the co-design of both the controller and the parameters of triggers. Moreover, to the best of the authors' knowledge, event-triggered  $H_\infty$  control for networked system considering the effect of network transmission delay has not been investigated. In order to demonstrate the effectiveness of our proposed method, a simulation example is finally given. In the example, some comparisons between some existing method and our method are also provided.

## II. THE SYSTEM DESCRIPTION AND AN EVENT GENERATOR

Consider the system described by

$$\dot{x}(t) = Ax(t) + Bu(t) + B_w w(t) \quad (1)$$

$$z(t) = Cx(t) + Du(t) \quad (2)$$

where  $x(t) \in \mathbb{R}^n$  and  $u(t) \in \mathbb{R}^m$  are the system state vector and the control input, respectively;  $w(t) \in \mathbb{R}^p$  is the external disturbance and  $w(t) \in L_2[0, \infty)$ ;  $z(t) \in \mathbb{R}^r$  is the objective vector;  $A$ ,  $B$ ,  $B_w$ ,  $C$  and  $D$  are the parameter matrices with appropriate dimensions.

Throughout this note, we assume that the system (1)–(2) is controlled through a network.

The purpose of this note is to design a linear controller  $u(t) = Kx(t)$ , where  $K$  is a gain matrix to be determined later, such that the resulting closed-loop system satisfies the required performance. In this note, the required performance is selected as an  $H_\infty$  performance. Different from a traditional point-to-point structure control system, the controller considered in this note is connected to the system (1)–(2) through a common digital communication network medium. In this case, the usual assumption that the sensor-controller-actuator communication is error-free, delay-free and without any limitation on the network bandwidth in a traditional point-to-point structure control system is no longer applicable. Under a periodic sampling mechanism, the  $H_\infty$  control design problem was investigated for (1)–(2) in [25] by considering the effect of the network transmission delay. However, how to save network resources such as network bandwidth was not considered in [25]. As is well known, in practical systems, a periodic sampling mechanism may often lead to the sending of many “unnecessary” signals through the network, which will turn to increase the load of network transmission and waste the network bandwidth, though this method has been widely used in many control systems. Therefore, it is significant to introduce a mechanism to determine which sampled signal should be sent out or not. In this section, a mechanism, called an event generator, as shown in Fig. 1, is constructed between the sensor and the controller which is used to determine whether the newly sampled state will be sent out to the controller by using the following judgement algorithm:

$$\begin{aligned} & [x((k+j)h) - x(kh)]^T \Omega [x((k+j)h) - x(kh)] \\ & \leq \sigma x^T((k+j)h) \Omega x((k+j)h) \end{aligned} \quad (3)$$

where  $\Omega$  is a symmetric positive definite matrix,  $j = 1, 2, \dots$ , and  $\sigma \in [0, 1)$ .

**Remark 1:** Notice that the sampled state  $x((k+j)h)$  satisfying the inequality (3) will not be transmitted. Only the one that exceeds the threshold in (3) will be sent to the controller, which means that, in the sensor side, only some of the sampled states that violate (3) will be sent out to the controller side. Obviously, compared with some existing method in [25], [29], [31], the burden of the network communication is reduced and the communication bandwidth in the network is saved.

In the wireless network, this method will also save the transmission

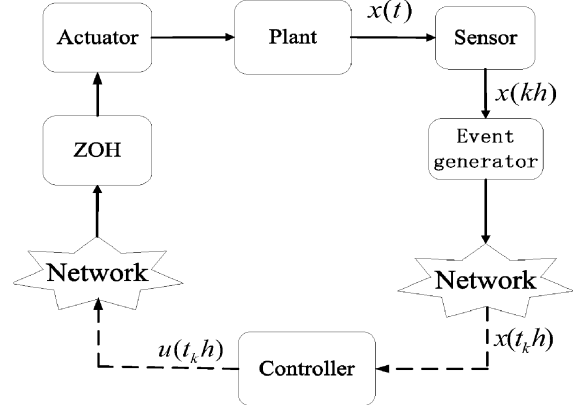


Fig. 1. Structure of an event-triggered networked control system.

energy, and then increase the lifespan of the battery of the nodes. Notice also that when  $\sigma = 0$ , the event-triggered scheme reduces to a periodic time-triggered scheme.

**Remark 2:** Different from a continuous event generator (CEG) which is an event generator with a continuous supervision of the state [10], [15]–[17], [19]–[21], the event generator with the algorithm (3) only supervises the difference between the states sampled in discrete instants having no interest in what happens in between updates. Moreover, unlike the CEG, the expensive work in retrofitting the existing system is also avoided because the detector, which can be implemented by a software, is constructed between the sensor and the controller.

Under the algorithm (3), assume that the release times are  $t_0 h, t_1 h, t_2 h, \dots$ , where  $t_0 = 0$  is the initial time.  $s_i h = t_{i+1} h - t_i h$  denotes the release period which corresponds to the sampling period given by the event generator in (3). For network uncertainties, we only consider the effect of the transmission delay on the system. Suppose that the time-varying delay in the network communication is  $\tau_k$  and  $\tau_k \in [0, \bar{\tau}]$ , where  $\bar{\tau}$  is a positive real number. Therefore, the states  $x(t_0 h), x(t_1 h), x(t_2 h), \dots$  will arrive at the controller side at the instants  $t_0 h + \tau_0, t_1 h + \tau_1, t_2 h + \tau_2, \dots$ , respectively.

**Remark 3:** The set of the release instants, i.e.,  $\{t_0, t_1, t_2, \dots\}$ , is a subset of  $\{0, 1, 2, \dots\}$ . The amount of  $\{t_0, t_1, t_2, \dots\}$  depends on not only the value of  $\sigma$ , but also the variation of the system state.

Based on the analysis above and considering the effect of the transmission delay, the system model under the event generator with (3) can be described as

$$\dot{x}(t) = Ax(t) + Bu(t_k h) + B_w w(t) \quad (4)$$

$$z(t) = Cx(t) + Du(t_k h), \quad t \in [t_k h + \tau_k, t_{k+1} h + \tau_{k+1}). \quad (5)$$

Under the control  $u(t) = Kx(t)$ , for  $t \in [t_k h + \tau_k, t_{k+1} h + \tau_{k+1})$ , (4)–(5) can be rewritten as

$$\dot{x}(t) = Ax(t) + BKx(t_k h) + B_w w(t) \quad (6)$$

$$z(t) = Cx(t) + DKx(t_k h). \quad (7)$$

We now consider the following two cases:

1) **Case A:** If  $t_k h + h + \bar{\tau} \geq t_{k+1} h + \tau_{k+1}$ , where  $\bar{\tau} = \max\{\tau_k\}$ , define a function  $\tau(t)$  as

$$\tau(t) = t - t_k h, \quad t \in [t_k h + \tau_k, t_{k+1} h + \tau_{k+1}). \quad (8)$$

Obviously

$$\tau_k \leq \tau(t) \leq (t_{k+1} - t_k) h + \tau_{k+1} \leq h + \bar{\tau}. \quad (9)$$

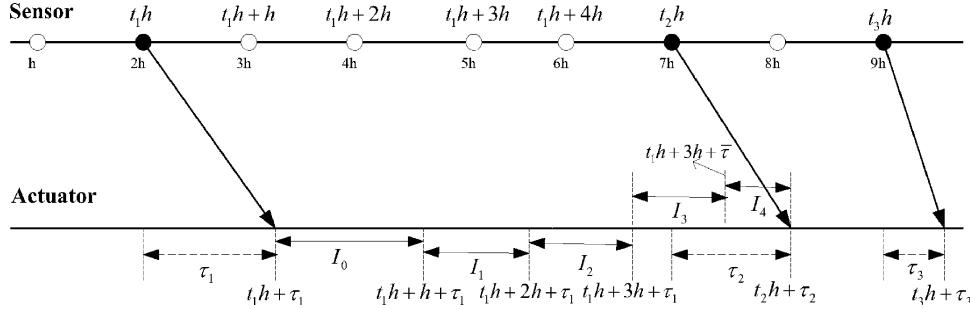


Fig. 2. Example of the partition of the interval and the sampling and release instants.

2) *Case B*: If  $t_k h + h + \bar{\tau} < t_{k+1} h + \tau_{k+1}$ , consider the following intervals

$$[t_k h + \tau_k, t_k h + h + \bar{\tau}), [t_k h + i h + \bar{\tau}, t_k h + i h + h + \bar{\tau}).$$

Since  $\tau_k \leq \bar{\tau}$ , it can be easily shown that  $d_M$  exists such that

$$t_k h + d_M h + \bar{\tau} < t_{k+1} h + \tau_{k+1} \leq t_k h + d_M h + h + \bar{\tau}$$

and  $x(t_k h)$  and  $x(t_k h + i h)$  with  $i = 1, 2, \dots, d_M$  satisfy (3). Let

$$\begin{cases} \mathcal{I}_0 = [t_k h + \tau_k, t_k h + h + \bar{\tau}), \\ \mathcal{I}_i = [t_k h + i h + \bar{\tau}, t_k h + i h + h + \bar{\tau}) \\ \mathcal{I}_{d_M} = [t_k h + d_M h + \bar{\tau}, t_{k+1} h + \tau_{k+1}) \end{cases} \quad (10)$$

where  $i = 1, 2, \dots, d_M - 1$ . One can see that

$$[t_k h + \tau_k, t_{k+1} h + \tau_{k+1}) = \cup_{i=0}^{d_M} \mathcal{I}_i. \quad (11)$$

In order to understand the partition of the interval in (11), we give an illustrative example in Fig. 2. Define

$$\tau(t) = \begin{cases} t - t_k h, & t \in \mathcal{I}_0 \\ t - t_k h - i h, & t \in \mathcal{I}_i, i = 1, \dots, d_M - 1 \\ t - t_k h - d_M h, & t \in \mathcal{I}_{d_M} \end{cases} \quad (12)$$

Then it can be easily shown that

$$\begin{cases} \tau_k \leq \tau(t) \leq h + \bar{\tau}, & t \in \mathcal{I}_0 \\ \tau_k \leq \bar{\tau} \leq \tau(t) \leq h + \bar{\tau}, & t \in \mathcal{I}_i, i = 1, \dots, d_M - 1 \\ \tau_k \leq \bar{\tau} \leq \tau(t) \leq h + \bar{\tau}, & t \in \mathcal{I}_{d_M} \end{cases} \quad (13)$$

where the third row in (13) holds because  $t_{k+1} h + \tau_{k+1} \leq t_k h + (d_M + 1) h + \bar{\tau}$ . Therefore, we have

$$0 \leq \tau_k \leq \tau(t) \leq h + \bar{\tau} \triangleq \tau_M, t \in [t_k h + \tau_k, t_{k+1} h + \tau_{k+1}). \quad (14)$$

In Case A, for  $t \in [t_k h + \tau_k, t_{k+1} h + \tau_{k+1})$ , define  $e_k(t) = 0$ . In Case B, define

$$e_k(t) = \begin{cases} 0, & t \in \mathcal{I}_0 \\ x(t_k h) - x(t_k h + i h), & t \in \mathcal{I}_i, i = 1, \dots, d_M - 1. \\ x(t_k h) - x(t_k h + d_M h), & t \in \mathcal{I}_{d_M} \end{cases} \quad (15)$$

**Remark 4:** Note that  $\tau_M = h + \bar{\tau}$ . When  $\tau_M$  is obtained from the following criteria in Section III, selecting a sampling period  $h < \tau_M$ , the allowable maximum transmission delay is  $\bar{\tau} = \tau_M - h$ . If  $\tau_k = 0$ , that is, no transmission delay exists or the effect of the transmission delay can be ignored, the obtained  $\tau_M$  denotes the maximum sampling period.

**Remark 5:** From the definition of  $e_k(t)$  and the triggering algorithm (3), it can be seen that for  $t \in [t_k h + \tau_k, t_{k+1} h + \tau_{k+1})$

$$e_k^T(t) \Omega e_k(t) \leq \sigma x^T(t - \tau(t)) \Omega x(t - \tau(t)). \quad (16)$$

Utilizing  $\tau(t)$  and  $e_k(t)$ , one can rewrite (6)-(7) as

$$\dot{x}(t) = A x(t) + B K x(t - \tau(t)) + B K e_k(t) + B_w w(t) \quad (17)$$

$$z(t) = C x(t) + D K x(t - \tau(t)) + D K e_k(t), \quad (18)$$

where  $t \in [t_k h + \tau_k, t_{k+1} h + \tau_{k+1})$ . For the system (17)-(18), we supplement the initial condition of the state  $x(t)$  on  $[-\tau_M, 0]$  as

$$x(t) = \phi(t), t \in [-\tau_M, 0] \quad (19)$$

where  $\phi(t)$  is a continuous function on  $[-\tau_M, 0]$ .

**Remark 6:** Notice that if  $\sigma = 0$ , then the corresponding  $e_k(t) = 0$ ; in this situation, the model described by (17)-(18) reduces to the one in [25]. Therefore, (17)-(18) can be “viewed” as a model for a networked control system with a time-triggered scheme and a perturbation which depends on the variation of the delayed state described by (16). Thus, one can see clearly that if  $\sigma \rightarrow 0^+$ , the dynamics of the system with an event-triggered scheme in this note will approach to the one with a time-triggered scheme.

### III. EVENT-TRIGGERED CONTROLLER DESIGN

In this section, for a given disturbance attenuation level  $\gamma$ , we design a state feedback controller such that the system described by (17)-(19) under the event generator with (3) meets the following two requirements.

- 1) The closed-loop system described by (17)-(19) with  $w(t) = 0$  is exponentially stable, which means that for any initial condition, there exist constants  $\epsilon > 0$  and  $\lambda > 0$  such that  $\|x(t)\| \leq \epsilon \sup_{-\tau_M \leq s \leq 0} \|\phi(s)\| e^{-\lambda t}$ ;
- 2) Under zero initial condition, the controlled output  $z(t)$  satisfies  $\|z(t)\|_2 \leq \gamma \|w(t)\|_2$  for any nonzero  $w(t) \in L_2[0, \infty)$ .

We now state and establish the following result.

**Theorem 1:** For given  $\gamma$ ,  $\sigma$  and matrix  $K$ , the system described by (17)-(19) is exponentially stable with an  $H_\infty$  norm bound  $\gamma$  if there exist matrices  $P > 0$ ,  $Q > 0$ ,  $R > 0$ ,  $\Omega > 0$ , and  $N, M$  with appropriate dimensions such that for  $l = 1, 2$

$$\begin{bmatrix} \Phi_{11} + \Gamma + \Gamma^T & * & * & * & * \\ \Phi_{21}^{(l)} & -R & * & * & * \\ \Phi_{31} & 0 & -\gamma^2 I & * & * \\ \Phi_{41} & 0 & \sqrt{\tau_M} R B_w & -R & * \\ \Phi_{51} & 0 & 0 & 0 & -I \end{bmatrix} < 0 \quad (20)$$

where  $\Gamma = [-N \quad N - M \quad M \quad 0]$  and

$$\Phi_{11} = \begin{bmatrix} P A + A^T P + Q & * & * & * \\ K^T B^T P & \sigma \Omega & * & * \\ 0 & 0 & -Q & * \\ K^T B^T P & 0 & 0 & -\Omega \end{bmatrix}$$

$$\begin{aligned}\Phi_{21}^{(1)} &= \sqrt{\tau_M} N^T, \quad \Phi_{21}^{(2)} = \sqrt{\tau_M} M^T \\ \Phi_{31} &= [B_\omega^T P \quad 0 \quad 0 \quad 0] \\ \Phi_{41} &= [\sqrt{\tau_M} R A \quad \sqrt{\tau_M} R B K \quad 0 \quad \sqrt{\tau_M} R B K] \\ \Phi_{51} &= [C \quad D K \quad 0 \quad D K].\end{aligned}$$

*Proof:* Choose the following Lyapunov functional as

$$V(t) = x^T(t) P x(t) + \int_{t-\tau_M}^t x^T(s) Q x(s) ds + \int_{t-\tau_M}^t \int_s^t \dot{x}^T(v) R \dot{x}(v) dv ds \quad (21)$$

where  $P, Q$ , and  $R$  are symmetric positive definite matrices. Employing (21) and introducing some free matrices [7], by using a similar method to the proof in [25] except for the treatment of the  $e_k(t)$ , we can conclude that the system described by (17)–(19) is exponentially stable with an  $H_\infty$  norm bound  $\gamma$  if (20) is satisfied. Due to page limitation, we omit the details in the proof.

*Remark 7:* From the definition of  $\tau(t)$ , it is clear that  $\tau(t)$  is a piecewise-linear function and satisfies  $\dot{\tau}(t) = 1$ . If this sawtooth structure characteristic of  $\tau(t)$  is taken into account, instead of the Lyapunov functional (21), we construct the following Lyapunov functional

$$\begin{aligned}\tilde{V}(t) &= x^T(t) P x(t) + \int_{t-\tau_M}^t x^T(s) Q x(s) ds \\ &+ \int_{t-\tau_M}^t \int_s^t \dot{x}^T(v) R \dot{x}(v) dv ds \\ &+ (\tau_M - \tau(t)) [x(t) - x(t - \tau(t))]^T Z_1 [x(t) - x(t - \tau(t))] \\ &+ (\tau_M - \tau(t)) \int_{t-\tau(t)}^t \dot{x}^T(s) Z_2 \dot{x}(s) ds.\end{aligned} \quad (22)$$

Then a less conservative result may be expected. However, due to page limitation, we do not list the result.

Based on Theorem 1, we are in a position to design the feedback gain  $K$  under the event trigger (3).

*Theorem 2:* For given  $\sigma, \gamma$  and  $\rho$ , the system described by (17)–(19) with the feedback gain  $K = YX^{-1}$  under the event trigger condition (3) is exponentially stable with an  $H_\infty$  norm bound  $\gamma$  if there exist matrices  $X > 0, \tilde{Q} > 0, \tilde{R} > 0, \tilde{\Omega} > 0$ , and matrices  $\tilde{N}, \tilde{M}, Y$  with appropriate dimensions such that  $l = 1, 2$

$$\begin{bmatrix} \Sigma_{11} + \tilde{\Gamma} + \tilde{\Gamma}^T & * & * & * & * \\ \Sigma_{21}^{(1)} & -\tilde{R} & * & * & * \\ \Sigma_{31} & 0 & -\gamma^2 I & * & * \\ \Sigma_{41} & 0 & \sqrt{\tau_M} B_\omega & \rho^2 \tilde{R} - 2\rho X & * \\ \Sigma_{51} & 0 & 0 & 0 & -I \end{bmatrix} < 0 \quad (23)$$

where  $\tilde{\Gamma} = [-\tilde{N} \quad \tilde{N} - \tilde{M} \quad \tilde{M} \quad 0]$  and

$$\Sigma_{11} = \begin{bmatrix} AX + XA^T + \tilde{Q} & * & * & * \\ Y^T B^T & \sigma \tilde{\Omega} & * & * \\ 0 & 0 & -\tilde{Q} & * \\ Y^T B^T & 0 & 0 & -\tilde{\Omega} \end{bmatrix}$$

$$\Sigma_{21}^{(1)} = \sqrt{\tau_M} \tilde{N}^T, \quad \Sigma_{21}^{(2)} = \sqrt{\tau_M} \tilde{M}^T$$

$$\Sigma_{31} = [B_\omega^T \quad 0 \quad 0 \quad 0]$$

$$\Sigma_{41} = [\sqrt{\tau_M} A X \quad \sqrt{\tau_M} B Y \quad 0 \quad \sqrt{\tau_M} B Y]$$

$$\Sigma_{51} = [C X \quad D Y \quad 0 \quad D Y].$$

*Remark 8:* From Theorem 2, one can see that for given  $\sigma, \gamma$  and  $\rho$ , the feedback gain  $K$  can be obtained by solving a set of LMIs in

(23). On the other hand, for the preselected  $\Omega$  and the feedback gain  $K$ , one can use Theorem 2 to derive the event triggered parameter  $\sigma$ . Therefore, one can employ Theorem 2 to co-design the feedback gain and the event trigger parameters.

*Remark 9:* Notice that the condition (23) includes information of the transmission delay. Thus, our method can be used to deal with the case with network transmission delay. For given condition on the transmission delay, by solving (23), the corresponding feedback gain and trigger parameters can be obtained, which can be used to guarantee the required performance even though the transmission delay exists in the network communication. On the other hand, for given  $\sigma, \rho$  and  $\gamma$ , the upper bound for  $\tau_M$  can be solved in terms of (23), and the larger  $\sigma$ , smaller upper bound of  $\tau_M$ . One can also see that a larger  $\sigma$  produces a larger average release period, which turns to reduce the load of network communication and thus decrease the transmission delay. This fact will be confirmed by a simulation example in the next section.

*Remark 10:* In [11] and [13], self-triggered control for a linear system is investigated and a technique is proposed to guarantee exponential input-to-state stability of the system. However, the system under consideration is free of delay and the method can not be applied to networked control systems. Moreover, the method in [11] and [13] can not be employed for co-design of both feedback gain and event trigger parameters.

*Remark 11:* The results obtained in this note can be easily extended to uncertain systems. Due to page limitation, the corresponding results are omitted.

*Remark 12:* It should be mentioned that the preliminary version of this note is published in [27]. Compared with [27], this note includes some new features for the proposed event-trigger scheme and derived criteria, see Remarks 6–9 in this note.

#### IV. AN EXAMPLE

Consider the system (1)–(2) with the parameter matrices in the following two cases:

1) *Case I:*  $C = B_w^T = [0 \quad 0 \quad 0 \quad 0]$ ,  $D = 0$  and

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{-mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{g}{l} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{Ml} \end{bmatrix}$$

where  $M = 10$  kg is the cart mass and  $m = 1$  kg is the mass of the pendulum bob,  $l = 3$  m is the length of the pendulum arm and  $g = 10$  m/s<sup>2</sup> is the gravitational acceleration. Since the eigenvalues of  $A$  are  $\{0, 0, 1.8257, -1.8257\}$ , the system is unstable without a controller. The state  $[x_1 \quad x_2 \quad x_3 \quad x_4] = [y \quad \dot{y} \quad \theta \quad \dot{\theta}]$ , where  $x_i$  ( $i = 1, 2, 3, 4$ ) are the cart's position, the cart's velocity, the pendulum bob's angle and the pendulum bob's angular velocity, respectively. The initial state is  $x_0 = [0.98 \quad 0 \quad 0.2 \quad 0]$ .

2) *Case II:* Matrices  $A$  and  $B$  are the same as the ones in Case I, the other parameter matrices are given as  $C = B_w^T = [1 \quad 1 \quad 1 \quad 1]$ ,  $D = 0.1$  and  $\omega(t) = \text{sgn}(\sin t)$  if  $t \in [0, 10]$ ; otherwise,  $\omega(t) = 0$ .

In the following, in Case I or Case II, we will demonstrate the design process of the feedback gain and the proposed event-triggering scheme and also provide some comparison results to show the advantage of our method over some existing ones.

In Case I, to compare with some existing methods in the design of an event-triggering scheme, we use the following feedback gain

$$K = [2 \quad 12 \quad 378 \quad 210] \quad (24)$$

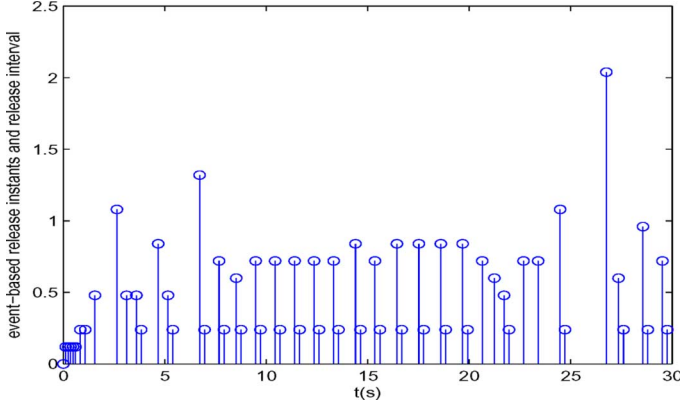


Fig. 3. Release instants and release interval in this note.

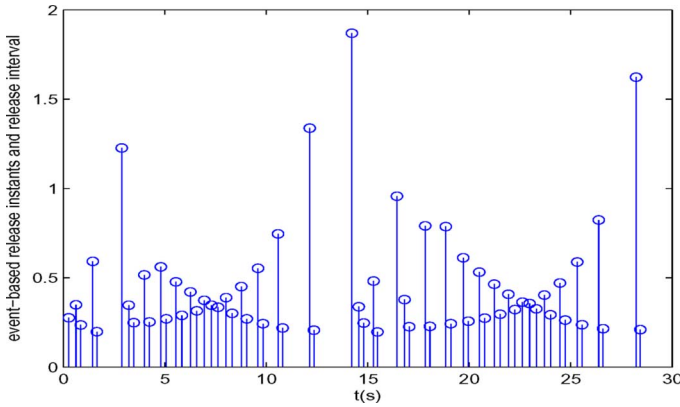


Fig. 4. Release instants and release interval using the event-triggering scheme in [19].

which is the same as the one in [21]. Applying Theorem 1 with  $\rho = 0.53$  and  $\sigma = 0.2$ , we obtain the maximum value of  $\tau_M$  as 0.12 and the corresponding  $\Omega$  as

$$\Omega = \begin{bmatrix} 0.0079 & 0.0310 & 1.0667 & 0.5466 \\ 0.0310 & 0.2247 & 5.8099 & 3.4684 \\ 1.0667 & 5.8099 & 191.3858 & 101.8792 \\ 0.5466 & 3.4684 & 101.8792 & 60.0510 \end{bmatrix}. \quad (25)$$

Suppose that  $\tau_k = 0$ . Since  $\tau_M = h + \bar{\tau}$ , it can be known that the maximum sampling period is 0.12. Taking  $h = 0.12$  and using the algorithm (3), the simulation results for  $t \in [0, 30]$  show that only 59 sampled signals need to be sent out to the controller, which takes 23.6% of the sampled signals. Moreover, it can be computed that our event generator leads to a maximum release interval of 2.04. The comparison results on the average interval under our scheme and other schemes in [3], [17], [19], [21] are shown in Table I. From this table, one can see that our event-triggering scheme can provide a larger average release interval than some existing ones in [3], [17], [19], [21]. We now take event-triggering schemes in this note and [19] as an example to show the release instants and release intervals, which are illustrated in Figs. 3 and 4. For the system (17)–(19) with the feedback gain (24), the state responses of our triggering scheme with the event-trigger matrix (25) and the event-triggering scheme in [19] are illustrated in Figs. 5 and 6, from which one can see that the steady state responses are at the same level.

For the given feedback gain (24), applying Theorem 1 with  $\sigma = 0$ , the upper bound of  $\tau_M$  is obtained as 0.1989. Suppose that  $\tau_k = 0$ . Let-

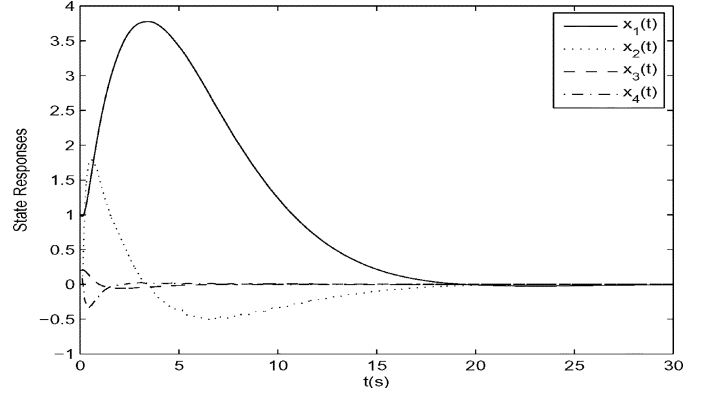


Fig. 5. State response using the event-triggered scheme with the trigger matrix (25).

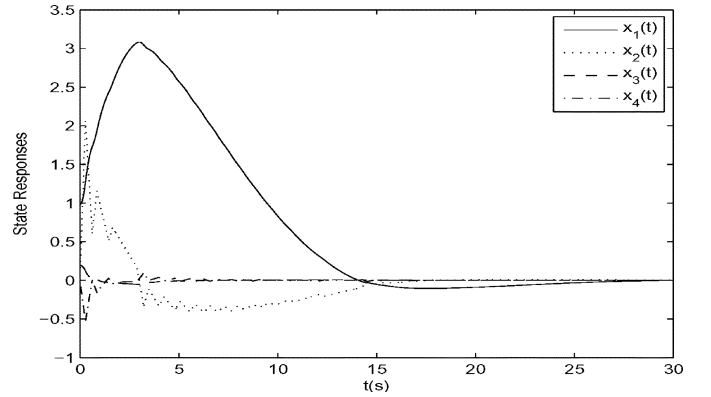


Fig. 6. State response using the event-triggered scheme in [19].

TABLE I  
AVERAGE SAMPLING PERIOD BY DIFFERENT TRIGGERING SCHEMES

| Event-triggering scheme                 | Average period |
|-----------------------------------------|----------------|
| Theorem 1 with $h = 0.12, \sigma = 0.2$ | 0.5131         |
| The scheme in [19]                      | 0.4816         |
| The scheme (Theorem 5.1) in [21]        | 0.3917         |
| The bound on MATI in [3]                | 0.0168         |
| The scheme in [17]                      | $< 10^{-5}$    |

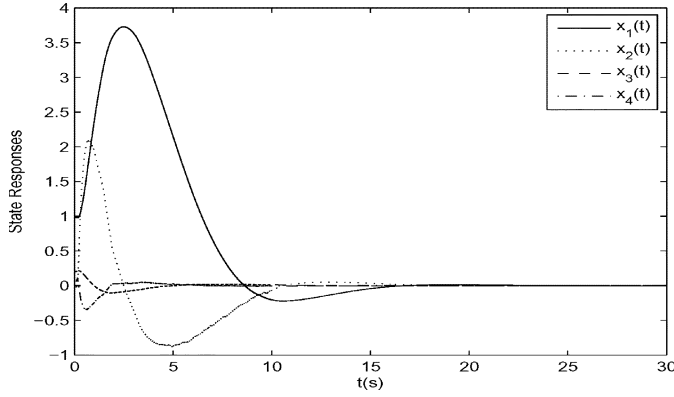
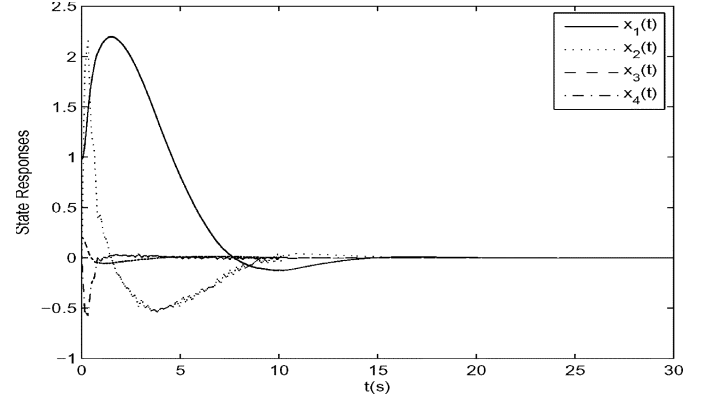
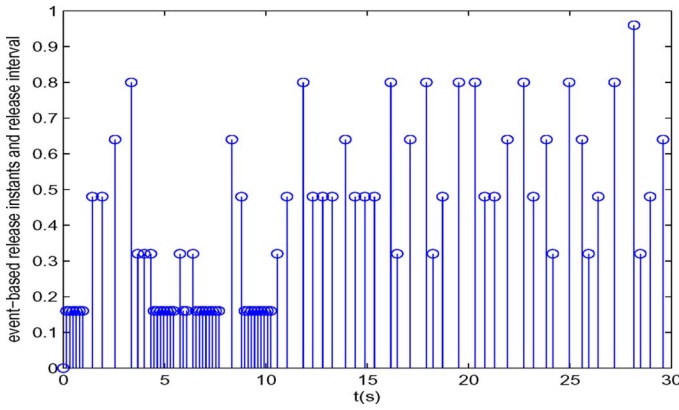
ting  $\bar{\tau} = 0$  and from  $\tau_M = h + \bar{\tau}$ , the maximum allowed sampling period is obtained as 0.1989. From Remark 1, when  $\sigma = 0$ , the event-triggered scheme reduces to the periodic release scheme. Compared 0.1989 with the result 0.5131 obtained above, the proposed event-triggered scheme can reduce much signal transmission and thus reduce the network burden than the periodic release scheme.

Applying Theorem 2 with  $\rho = 0.53$  and  $\sigma = 0.2$ , the upper bound for  $\tau_M$  is computed as 0.16 and the corresponding feedback gain and the trigger matrix are

$$K = [0.7823 \quad 5.8233 \quad 275.0062 \quad 152.2533] \quad (26)$$

$$\Omega = \begin{bmatrix} 0.0095 & 0.0564 & 2.5577 & 1.4152 \\ 0.0564 & 0.4463 & 18.8120 & 10.4375 \\ 2.5577 & 18.8120 & 881.5504 & 488.0012 \\ 1.4152 & 10.4375 & 488.0012 & 270.2932 \end{bmatrix}. \quad (27)$$

If  $h$  is selected as 0.01, under the controller with feedback gain (26) and the trigger (3) with  $\Omega$  in (27), the closed-loop system can tolerate the transmission delay bounded by 0.15.

Fig. 7. State responses for  $h = 0.16$ ,  $K$  and  $\Omega$  in (28) and (29).Fig. 9. State responses for  $h = 0.06$  and  $\tau_M = 0.1$ .Fig. 8. Release instants and release interval for  $h = 0.16$ ,  $K$  and  $\Omega$  in (28) and (29).

In Case II, using Theorem 2 with  $\rho = 0.53$ ,  $\gamma = 200$  and  $\sigma = 0.1$ , we obtain the upper bound of  $\tau_M$  as 0.16, the feedback gain  $K$  and the event-triggering matrix  $\Omega$  as

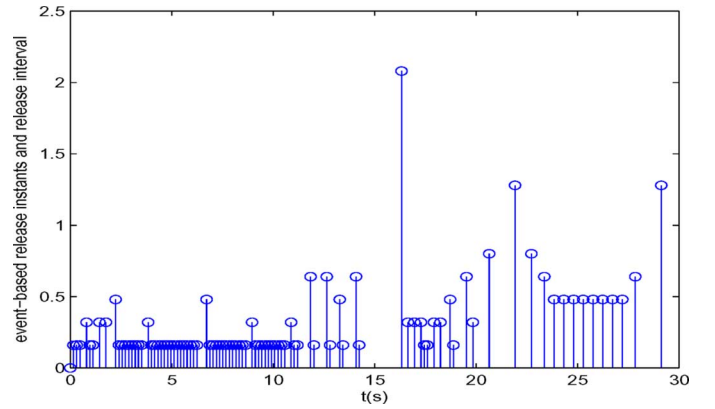
$$K = \begin{bmatrix} 2.9129 & 10.4357 & 287.9029 & 160.3271 \end{bmatrix} \quad (28)$$

$$\Omega = \begin{bmatrix} 5 & 16 & 427 & 238 \\ 16 & 57 & 1522 & 848 \\ 427 & 1522 & 41774 & 23263 \\ 238 & 848 & 23263 & 12954 \end{bmatrix}. \quad (29)$$

Setting  $h = 0.16$  and  $\tau_k = 0$ , the average release period is obtained as 0.3844, which is larger than 0.2830 obtained in [21]. The state response and release instants and release interval of the system with  $h = 0.16$ ,  $K$  and  $\Omega$  in (28) and (29) are shown in Figs. 7 and 8.

If selecting the sampling period  $h = 0.06$ , since  $\tau_M = h + \bar{\tau}$ , the corresponding allowable maximum transmission delay is 0.1. Considering the effect of the transmission delay, the average release period is obtained as 0.3098, the state response and release instants and release interval of the system with  $h = 0.06$ ,  $\tau_M = 0.1$  and  $K$ ,  $\Omega$  in (28) and (29) are shown in Figs. 9 and 10.

*Remark 13:* It should be pointed out that in [27], the computation result was incorrect. This mistake is corrected in this note. Furthermore, compared with [27], this note compares with some existing results on the steady state responses are made in order to show the proposed event-trigger scheme can reduce the communication burden when maintaining the same level on the state response.

Fig. 10. Release instants and release interval for  $h = 0.06$  and  $\tau_M = 0.1$ .

## V. CONCLUSION

A novel event-triggering scheme has been proposed to reduce the communication load in the network and this scheme can be used to determine when the sampled signals by sensors will be transmitted. Under this event trigger scheme, an event-triggered  $H_\infty$  control design method has been developed for networked control systems with network-induced delay. A delay system model has been utilized to describe the prosperities of the event trigger and the effect of the transmission delay on the system. Criteria for stability with an  $H_\infty$  norm bound and  $H_\infty$  control design have been derived, which are expressed in the form of linear matrix inequalities. A simulation example has shown that our event-triggering scheme can lead to a larger average release period than some existing schemes.

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## Stable Interval Observers in $\mathbb{C}$ for Linear Systems With Time-Varying Input Bounds

Christophe Combastel

**Abstract**—This technical note deals with the design of stable interval observers and estimators for continuous-time linear dynamic systems under uncertain initial states and uncertain inputs enclosed within time-varying zonotopic bounds. No monotony assumption such as cooperativity is required in the vector field: the interval observer stability directly derives from the stability of the observer state matrix, where any poles (real or complex, single or multiple) are handled in the same way.

**Index Terms**—Error estimation, fault diagnosis, intervals, observers, robustness, stability, state estimation, verification.

### I. INTRODUCTION

Computing outer approximations of the states of uncertain dynamic systems appears to be a key issue in many contexts where automated decision-making is required. Indeed, a single point value or estimate alone is not suitable to make a robust decision: to that purpose, the point value should be compared to some estimation of its admissible range according to the type of decision to be taken. In particular, both fault diagnosis and verified model-based design require computationally effective algorithms supporting robustness analysis in order to perform logically sound detection and isolation of inconsistencies.

This technical note focuses on the computation of guaranteed bounds for the state estimates  $x(t) \in \mathbb{C}^n$  at time  $t \in \mathbb{R}^+$  defined by complex-valued observers as in (1)

$$\dot{x}(t) = Ax(t) + \phi(t) \quad (1)$$

$$x(0) \in [x^-(0), x^+(0)] \quad (2)$$

$$\phi(t) = u(t) + Z(t)s(t), \quad \forall t, s(t) \in [-1, +1]^p \subset \mathbb{R}^p \quad (3)$$

where  $A \in \mathbb{C}^{n \times n}$  is a strictly stable known matrix (i.e., all its eigenvalues have a strictly negative real part). The initial state estimate  $x(0) \in \mathbb{C}^n$  and the input term  $\phi(t) \in \mathbb{C}^n$  are assumed to be unknown but bounded within known bounds.  $x(0) \in \mathbb{C}^n$  is assumed to belong to an interval vector defined by inferior and superior vector bounds (2):  $x(0) \geq x^-(0) \in \mathbb{C}^n$ ,  $x(0) \leq x^+(0) \in \mathbb{C}^n$ , and  $x^-(0) \leq x^+(0)$  where the inequality operator should be understood element-by-element (including real and imaginary parts, as it will be more precisely stated in Section IV).  $\phi(t)$  is assumed to belong to a time-varying compact set defined as in (3), where  $u(t) \in \mathbb{C}^n$  is a known vector and  $Z(t) \in \mathbb{C}^{n \times p}$  is a known matrix.  $\phi(t)$  is thus bounded by a time-varying zonotope [1]. This class of centrally symmetric polytopic domains is rather versatile while leading to computationally effective implementations of some basic operators, e.g., linear image, support function. In the particular case where  $Z(t)$  is a square diagonal matrix,  $\phi(t)$  is bounded by an interval vector at each time  $t$ . Some regularity conditions are also introduced: as each scalar element of  $u(t)$ ,  $Z(t)$  and  $s(t)$  is assumed to be a continuous function of time, so is  $\phi(t)$ .

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